

BUILDING MATERIALS

➤ STONES

➤ **BRICKS**

➤ CEMENT

➤ CONCRETE

➤ STEEL

Bricks are artificial blocks manufactured from tempered clay into standard sizes.



Bricks - Introduction

- Sun-dried bricks - With the utilization of fire became burnt bricks
- In Mesopotamia, palaces and temples were built of stone and sun-dried bricks in 4000 B.C.
- Invention of kilns made mass production of bricks easy
- Romans made the first large-scale use of masonry arches and roof vaults in their basilica (domes), baths and aqueducts (arch bridge)



Uses of bricks



So on....

Chemical composition of brick

Ingredients	Concentration (%)
Silica	50 – 60
Alumina	20 – 30
Iron oxide	5 – 6
Lime	<5
Magnesia	0.1

The clay should be free from pebbles, gravel, alkalis, organic matter, vegetation and iron pyrites.



Constituents of good brick

- **Silica** - A good brick earth should contain about 50 to 60 percent of silica. Presence of silica prevents cracking, shrinking and warping of raw bricks. It thus imparts uniform shape to the bricks. Durability of bricks depends on the proper proportion of silica in brick earth. Excess of silica destroys the cohesion between particles and bricks become brittle.
- **Alumina** - It is the chief constituent of every kind of clay. A good brick earth should contain 20 to 30 percent of alumina. This constituent imparts plasticity to earth so that it can be moulded. If alumina is present in excess, raw bricks shrink and warp during drying and burning.
- **Oxide of iron** - A small quantity of oxide of Iron to the extent of 5 to 6 percent is desirable in good brick to imparts red colour to bricks. Excess of oxide of iron makes the bricks dark blue or blackish.
- **Lime** - A small quantity of lime is desirable in finely powdered state to prevents shrinkage of raw bricks. Excess of lime causes the brick to melt and hence, its shape is lost due to the splitting of bricks.
- **Magnesia** - A small quantity of magnesia in brick earth imparts yellow tint to bricks, and decreases shrinkage. But excess of magnesia decreases shrink leads to the decay of bricks.

Manufacture of clay bricks

Preparation

- **Process** (digging, crushing, sieving, grinding, proportioning, mixing and tempering)
- **Add chemicals** for special purpose (barium carbonat, soluble salts)

Formation

- Process of giving shape to the bricks.
- Variety shaping methods depends on moisture content and consistency of clay
- Ex: Semi dry process, stiff plastic process, wire cut process, soft mud process.

Drying

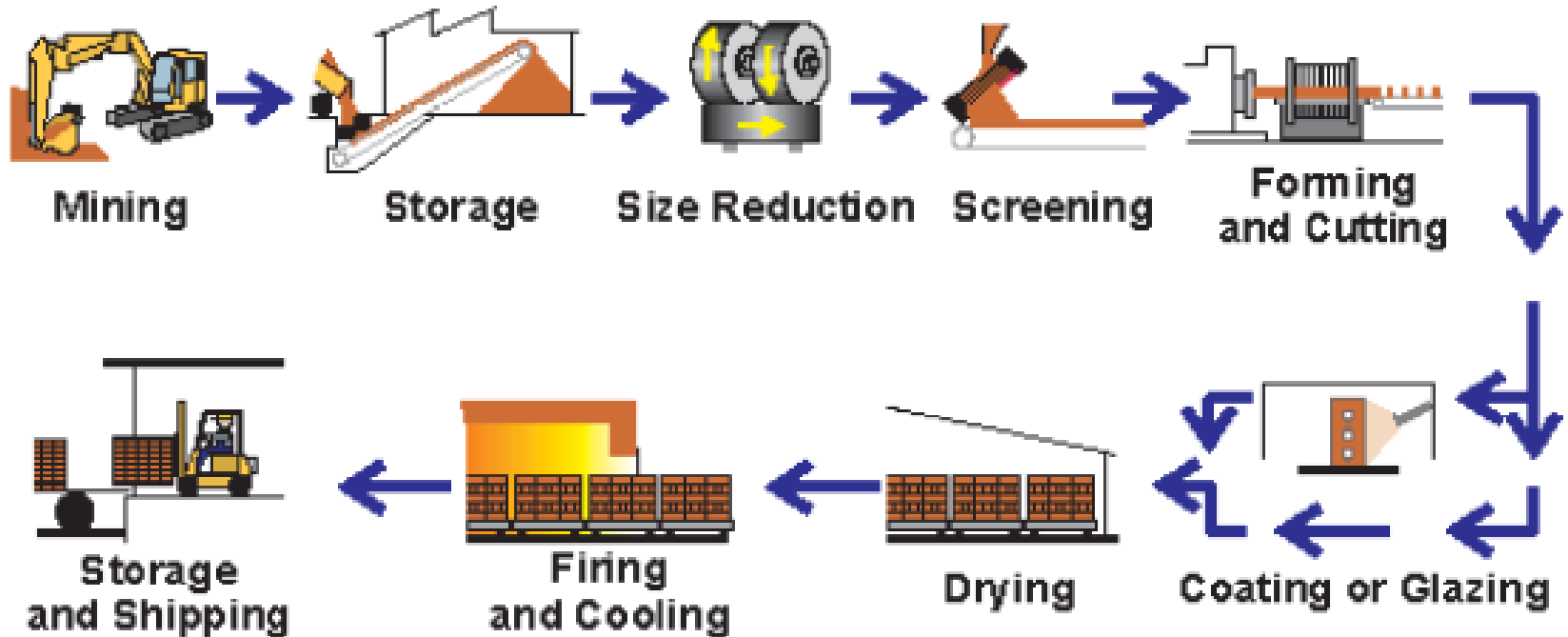
- To reduce the moisture content
- Enable brick to be stacked higher in the kiln
- Enable firing temperature to be increased more rapidly

Firing

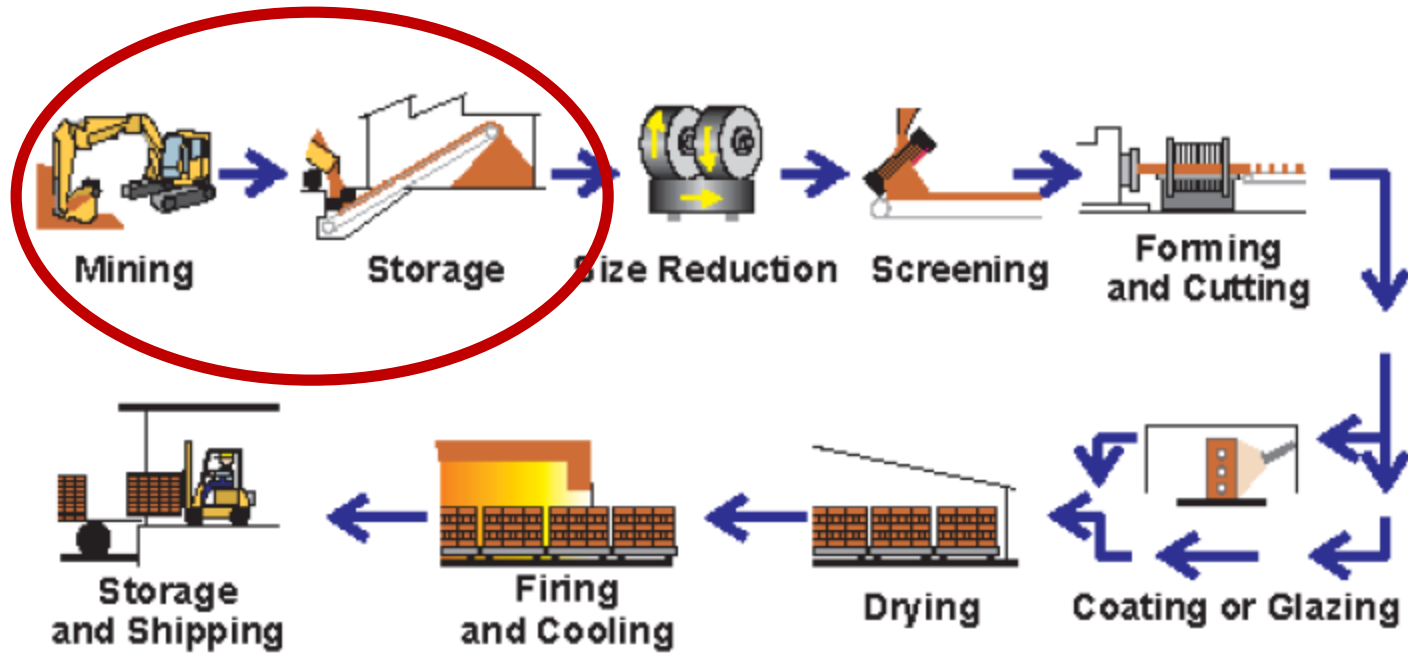
- Changes the physical structure and give good mechanical properties and resistance to water



Brick Manufacturing process



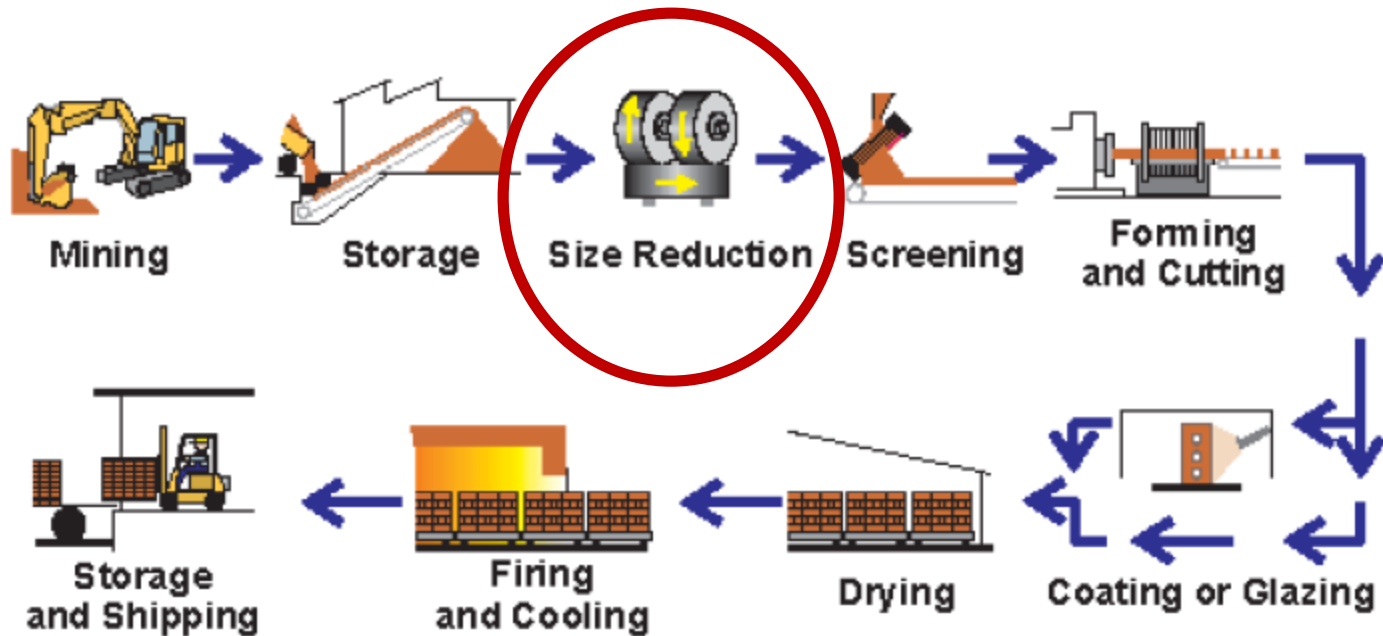
Manufacture Process Overview



Mining and storage

- **Removal of loose soil:** The top layer of ground should be removed for a depth of 200 mm. It should not be used for preparation of bricks as it contains lot of impurities.
- **Digging, spreading and cleaning:** The earth removed from below 200 mm is spread on a level ground. All undesirable materials like grass, roots, gravel etc. are removed.

Manufacture Process Overview

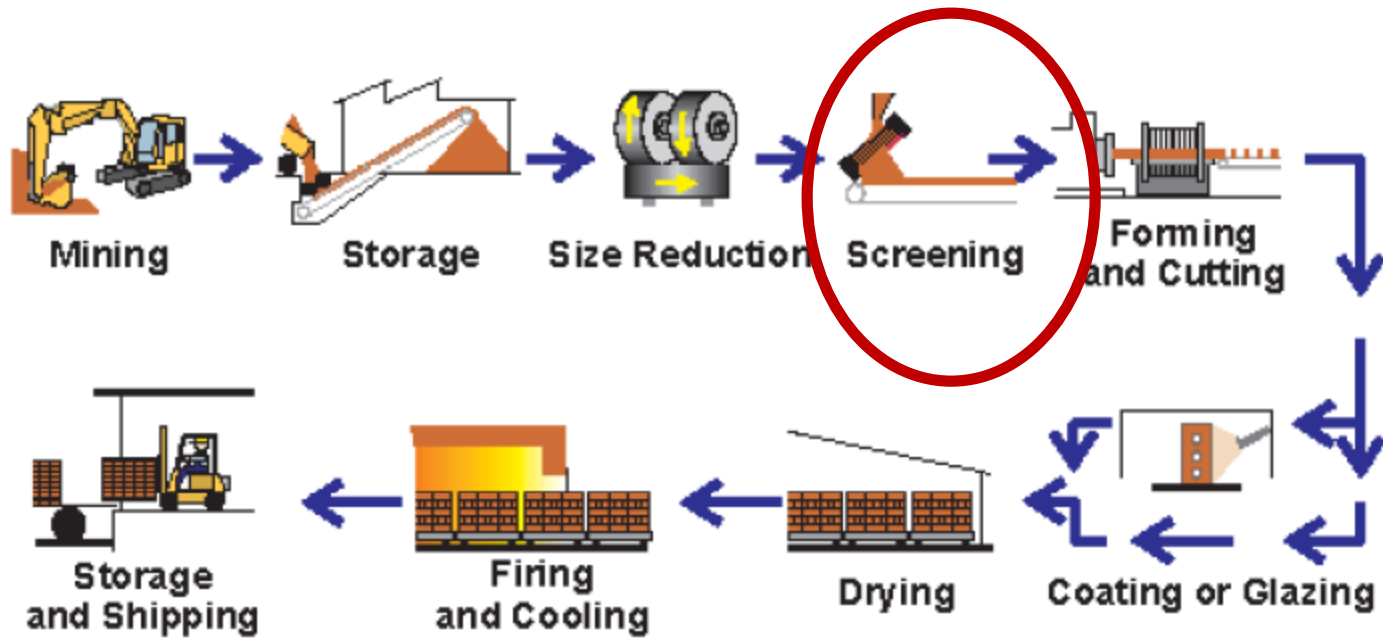


Size Reduction

- Dry Pan Crushers- Breaks large chunks of clay or shale
- Grinders- pulverizes the clay to a fine consistency (Muller Wheels)
- Clays should be dry when grinding

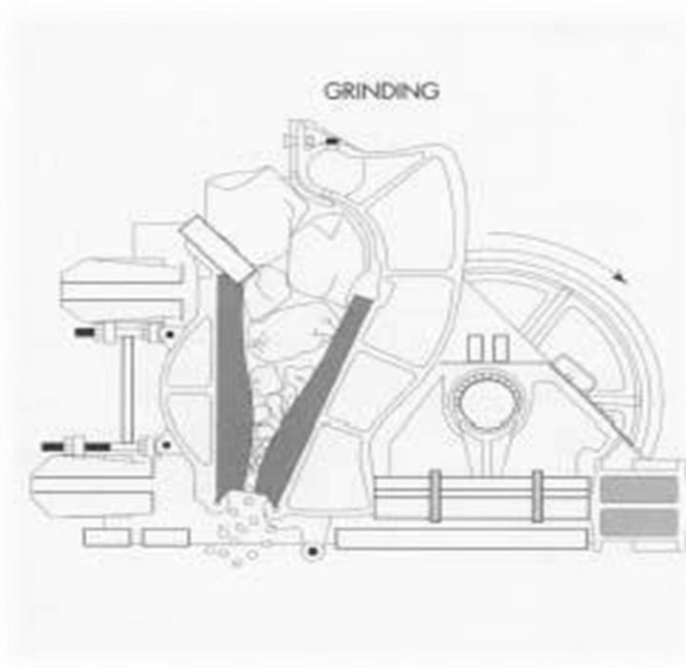


Manufacture Process Overview

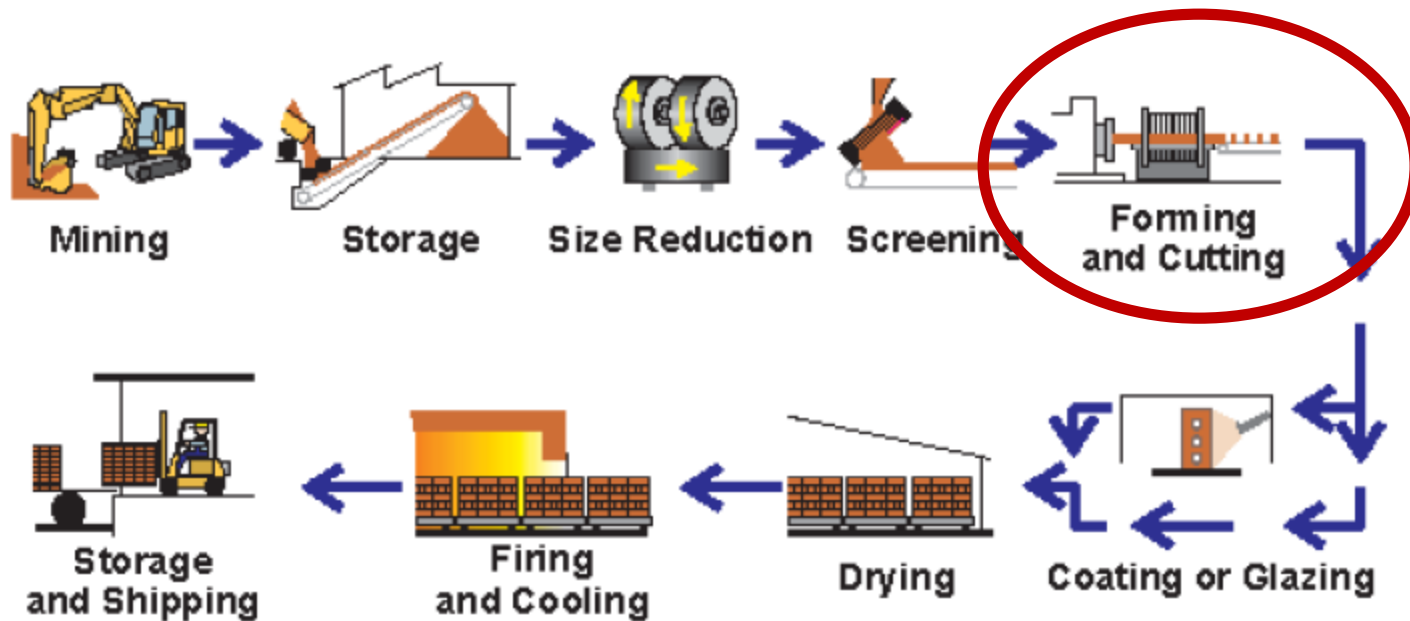


Screening

- Passes through Vibrating Screen, Scalping Screen
- Coarse material left goes back to grinder - Hammer mill
- Large particles inhibit the durability of the brick



Manufacture Process Overview

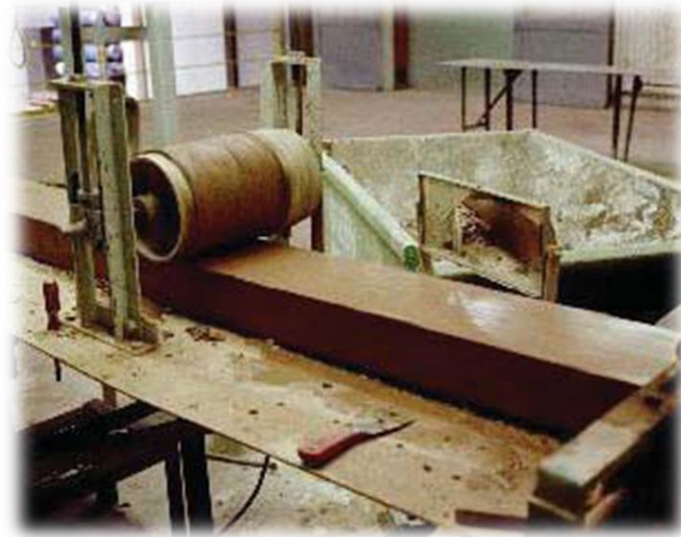


Forming

➤ Two steps in the forming process

✓ Tempering

- This is done to make the whole mass of clay, homogenous and plastic.
- This is done by trampling under feet by men, cattle in a Pug mill or by mixer when bricks are required on large scale.

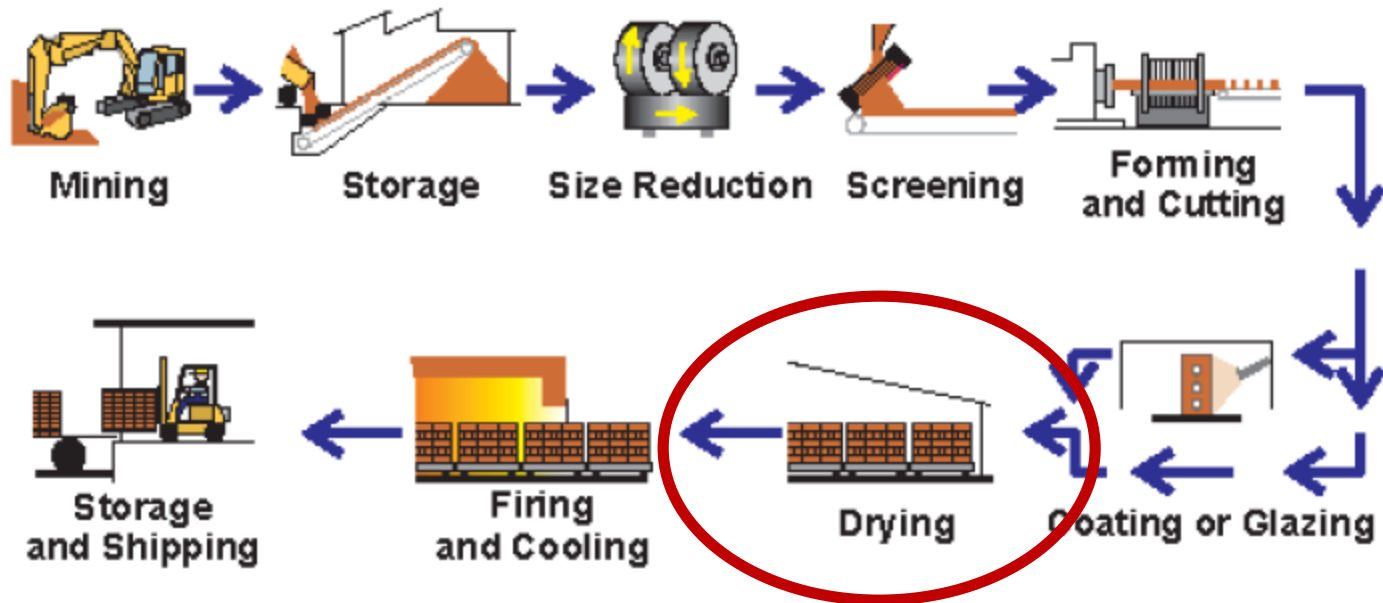


✓ Molding

- The pugged brick earth is then moulded into shape in moulds.
- Moulding can be done on the ground or on top of a table, either by hand or machine.

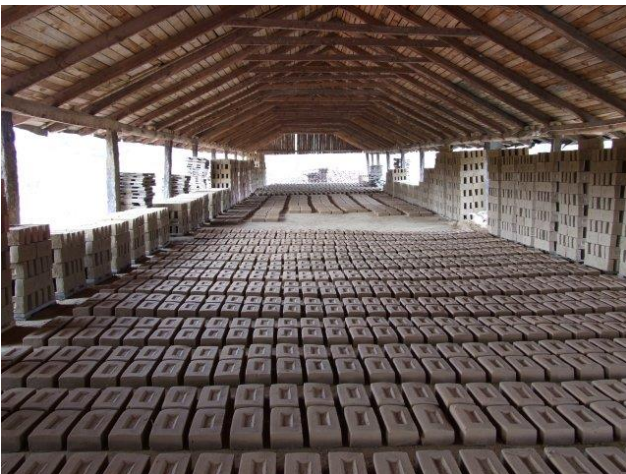


Manufacture Process Overview

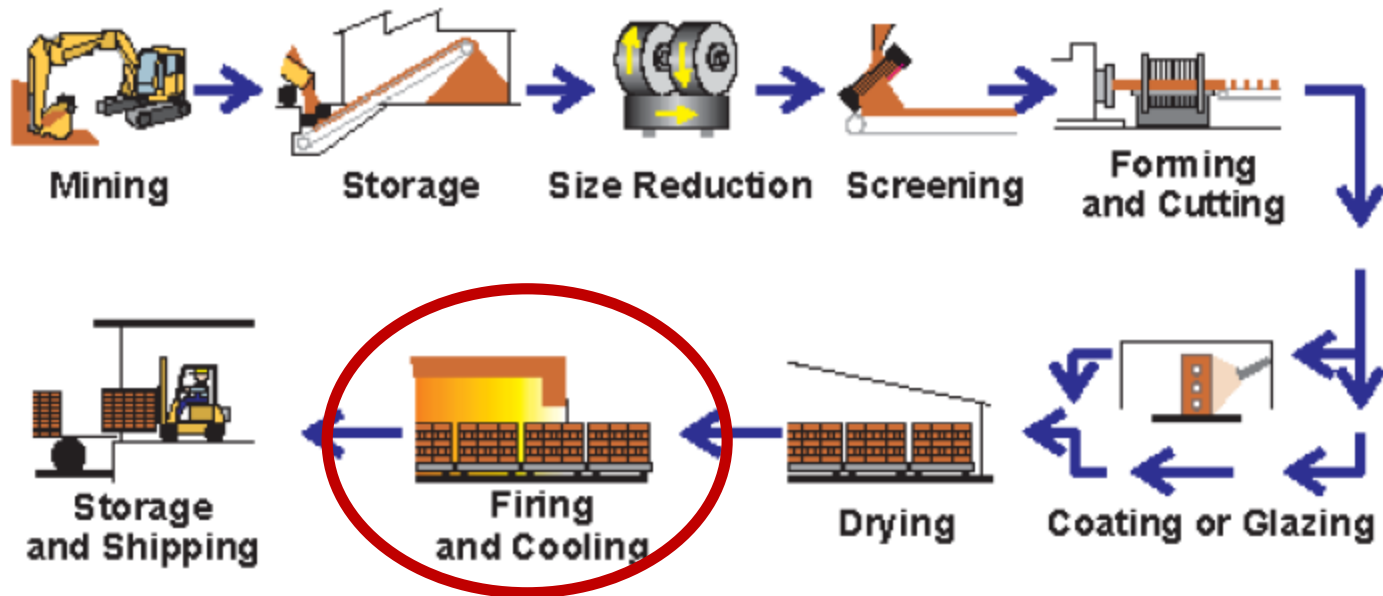


Drying

- In artificial drying method using oven, it takes < 48 hours.
- The moisture content of the formed clay is reduced, not eliminated, so that the brick will not explode during the firing process.
- In natural drying method, bricks are dried for a period of 7 to 14 days. During drying this must be protected from wind, rain and direct sun.



Manufacture Process Overview

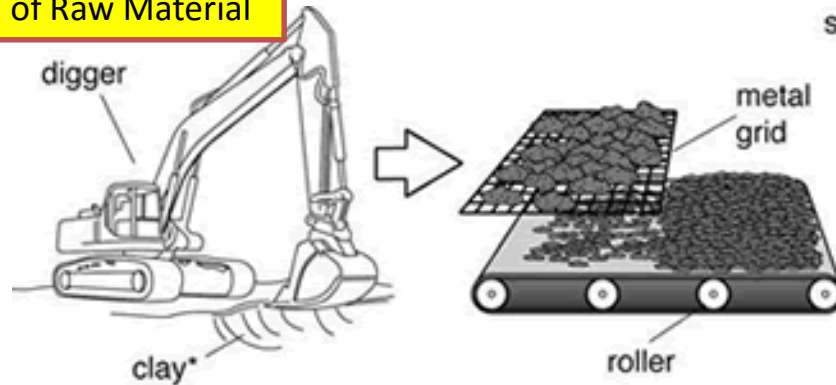


Firing and cooling

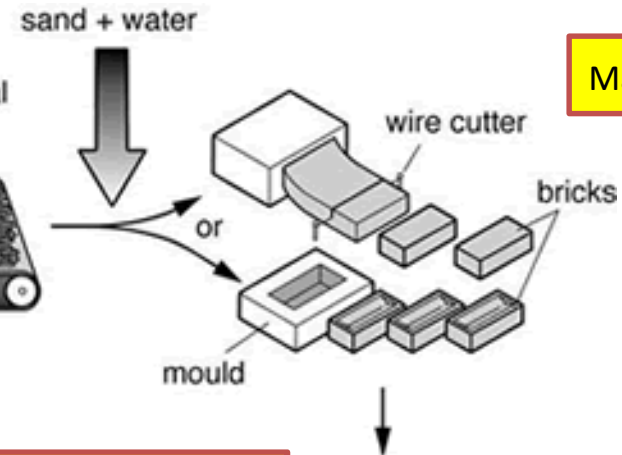
- The dried brick is sent through a firing chamber for 5 to 10 hours at 200 to 900°C.
- Heat and humidity must be regulated to avoid cracking in the brick.
- Later, the temperature will be ramped to 1300°C for 10 hours
- The bricks are allowed to cool for 48 – 72 hours
- Later the bricks will be stocked for delivery.

Brick manufacturing with advanced technology

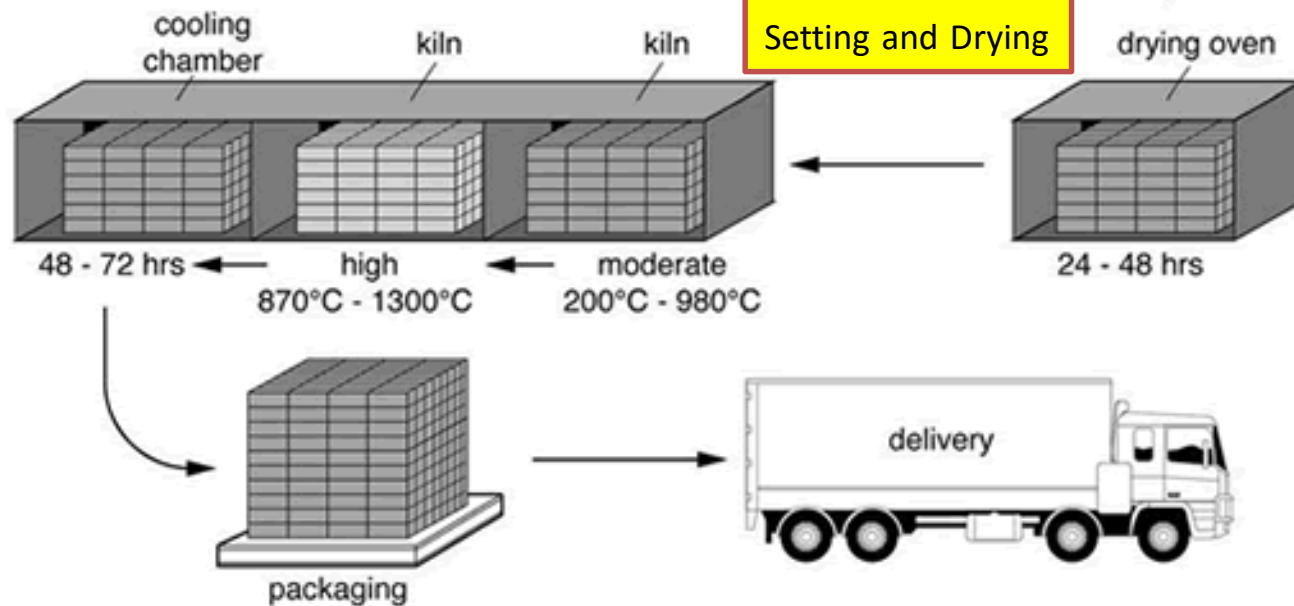
Preparation of Raw Material



Making Process



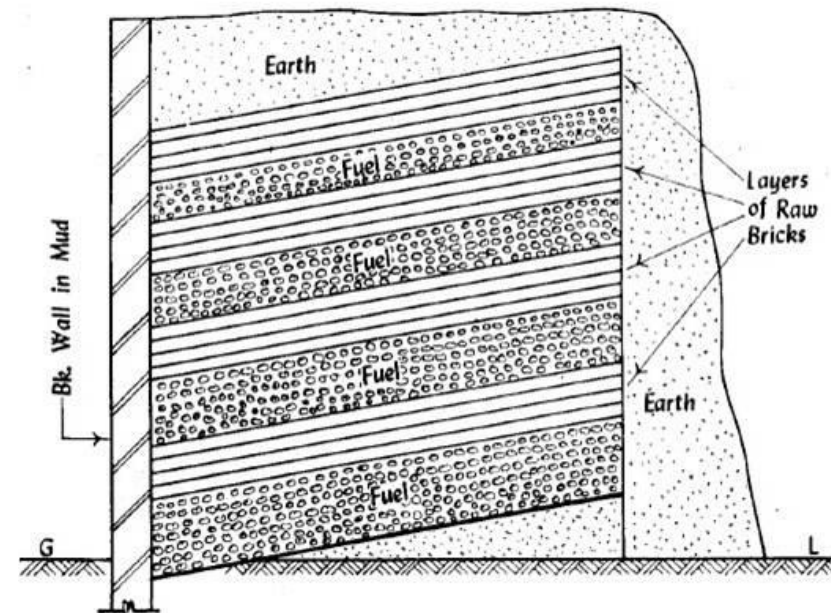
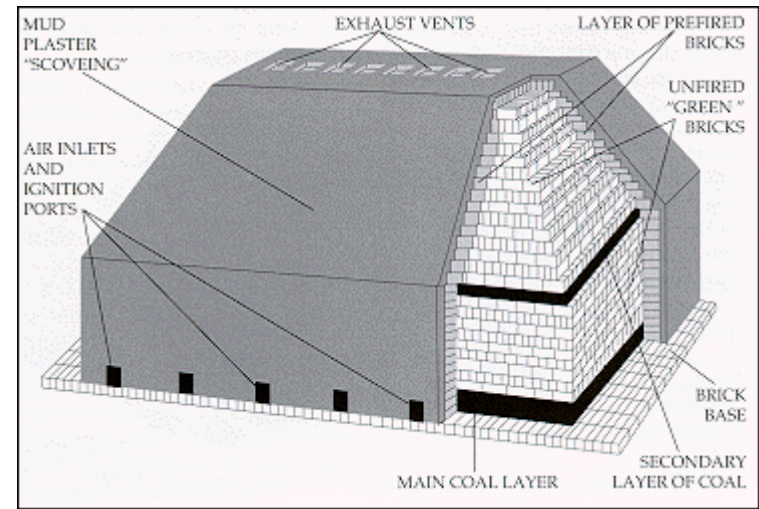
Setting and Drying



Types of firing of bricks

Burning in Clamps:

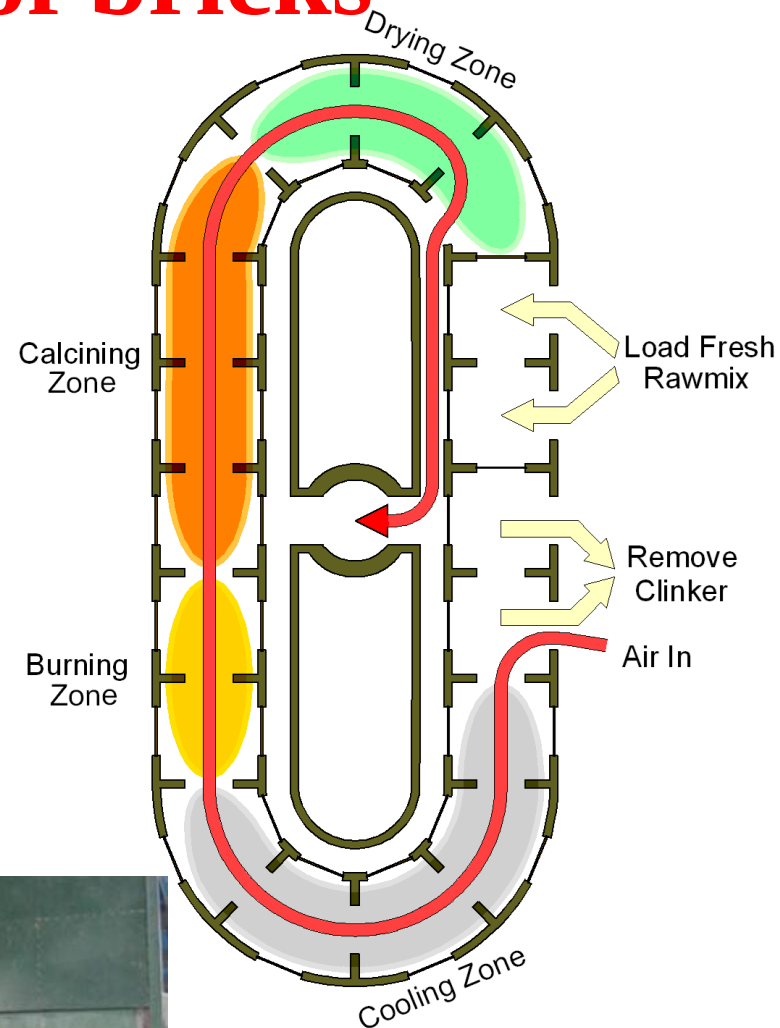
- A clamp is an arrangement where the bricks and fuel are arranged in alternate layers.
- A floor is prepared and the fuel is first spread at slope of 15° . The fuel consists of wooden piece, cow dung, straw, waste paper etc., On top of this layer the raw bricks are placed.
- Then another layer, of 75 cm. thick fuel is spread. Thus alternate layer of brick and fuel are arranged to a height of 3 to 4 m.
- Then entire clamp is plastered on all the sides and top and filled with clay to conserve the heat.
- The bricks are burnt for a period of one or two months and then cooled for the same period.



Types of firing of bricks

Burning in Kilns:

- When large number of bricks have to be burnt, Kilns are adopted.
- Kilns are of two types
 - Intermittent Kilns and
 - Continuous Kilns
- In intermittent Kilns, the loading firing cooling and unloading are done one after the other and not simultaneously,
- In continuous kilns, these operations are carried out without any interruptions.



Merits and demerits of Clamp and Kiln burning

➤ **Clamp Burning:**

- ✓ Only 60% of the bricks will be of first class variety.
- ✓ Bricks can be used for temporary structure
- ✓ Initial cost is low
- ✓ Amount of fuel used is low and hence economical
- ✓ Continuous supervision is not necessary
- ✓ Quality of bricks is poor.
- ✓ Firing cannot be regulated.
- ✓ It takes a long time for burning and cooling the bricks (3 to 4 months).
- ✓ There is large amount of wastage of heat from the hot gases.

Merits and demerits of Clamp and Kiln burning

➤ Kiln Burning:

- ✓ Quality of bricks turned out is good. 90% first class bricks
- ✓ Bricks can be used for permanent structure
- ✓ Time of firing is only 24 hours and cooling is 4 days.
- ✓ It is yielding about 25,000 bricks per day
- ✓ Heat of hot gases is utilized in heating bricks to be burnt.
- ✓ Initial cost is higher
- ✓ Fuel cost is higher
- ✓ Constant supervision is necessary

Types of bricks

- Classification based on dimensions
 - ✓ Conventional / Traditional bricks: Size: 230 mm X 114 mm X 76 mm
 - ✓ Standard / Modular : Size : 190 mm X 90 mm X 90 mm
- Classification based on their manufacturing method:
 - ✓ Hand moulded (ground or table moulded)
 - ✓ Machine moulded (wire cut)
- Classification based on their use:
 - ✓ Building (common) brick
 - ✓ Facing brick
 - ✓ Floor / paving brick
 - ✓ Furnace brick
- Classification based on their quality for a special utility:
 - ✓ 1st class (grade A)
 - ✓ 2nd class (grade B)
 - ✓ 3rd class (grade C)
 - ✓ 4th class (grade D)

Classification of Bricks based on its quality

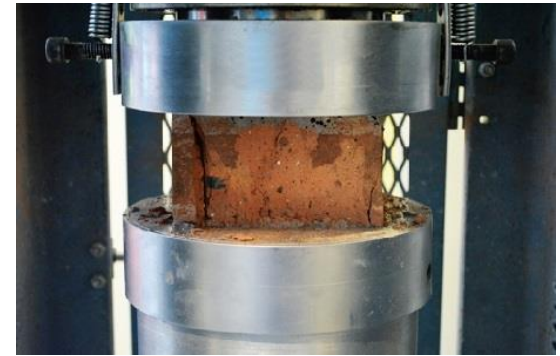
- 1. First class bricks:** Burnt in kilns - Table-moulded - Have standard shape - Surfaces and edges of the bricks are sharp, square, smooth and straight - Used for superior work of permanent nature.
- 2. Second class bricks:** Burnt in kilns - Ground-moulded - Surface is somewhat rough and shape is also slightly irregular - Have hair cracks - Edges may not be sharp and uniform - Used at places where brickwork is to be provided with a coat of plaster
- 3. Third class bricks:** Ground-moulded - Not hard - Rough surfaces with irregular and distorted edges - Dull sound when struck together - Used for unimportant and temporary structures and at places where rainfall is not heavy.
- 4. Fourth class bricks:** Over burnt - Irregular shape - Dark color -- As they are over burnt, sometimes found to be stronger and compact structure, than that of first class bricks - Used as aggregate for concrete in foundations, floors, roads, etc.

Mechanical properties of brick

- Compressive/crushing strength – *resistance to load*
- Soundness – *clear ringing sound*
- Hardness – *nail impression*
- Water absorption – *% change in weight*
- Linear shrinkage – *% change in length*
- Efflorescence – *deposit of salt*
- Thermal conductivity
- Fire resistance
- Sound insulation

Testing of bricks

1. **Crushing strength test:** Minimum crushing strength of bricks is 3.5 N/mm^2 and for superior bricks; it may vary from 7 to 14 N/mm^2
2. **Soundness test:** Two bricks are taken and they are struck with each other. Bricks should not break and a clear ringing sound should be produced.
3. **Hardness test:** A scratch is made on brick surface with the help of a finger nail. If no impression is left on the surface then the brick is treated to be sufficiently hard.



Testing of bricks (Contd.)

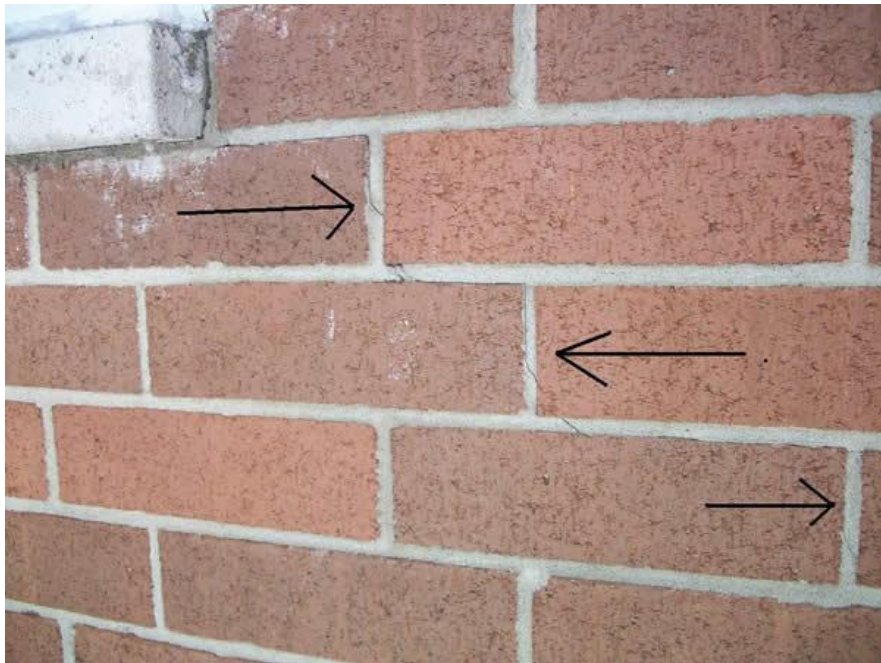
4. Absorption: Weight of water absorbed after 24 hours in cold water divided by the dry unit weight. A good brick should **absorb less than 20% of water** by weight. Water absorption affects the durability of brick. **Hard bricks** absorb less than **10% water**.

Very dry bricks will have high suction values, and should be thoroughly wetted 3 - 24 hours prior to use, to prevent excessive drying of the mortar before hydration.



Testing of bricks (Contd.)

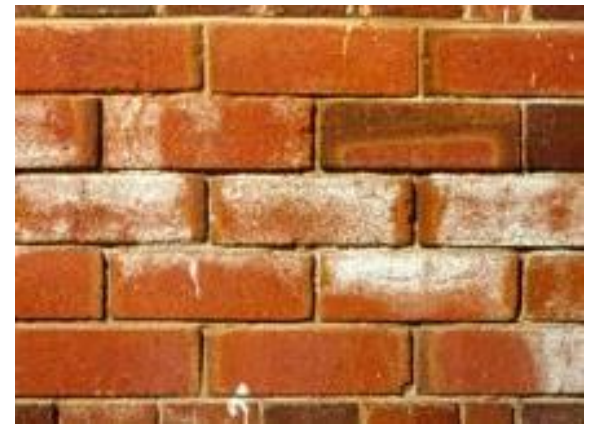
5. Linear shrinkage is the change in length of a brick from a wet to a dry condition, and is limited to 0.065%. The amount of shrinkage depends on the moisture content in the brick and the humidity. The greater the moisture content, the higher the shrinkage when drying. Since, bricks should be wet prior to construction, shrinkage will lead to cracks at the mortar joints



Testing of bricks (Contd.)

6. **Efflorescence** is the migration (leaching) of white crystals (salts) to the surface of the brick work.

- ✓ For finding out the presence of soluble salts in a brick, it is immersed in water for 24 hours and allowed to dry in shade.
- ✓ Presence of grey or white deposits on the surface of brick
 - ~ 10 %, the efflorescence is slight and considered as slight or moderate,
 - ~ 50 %, the efflorescence becomes heavy and it is treated as serious, when such deposits are converted into powdery mass.



- ✓ In real structures, it occurs in wet-dry cycles.
- ✓ To prevent efflorescence, walls should be kept as dry as possible, and good drainage should be provided.

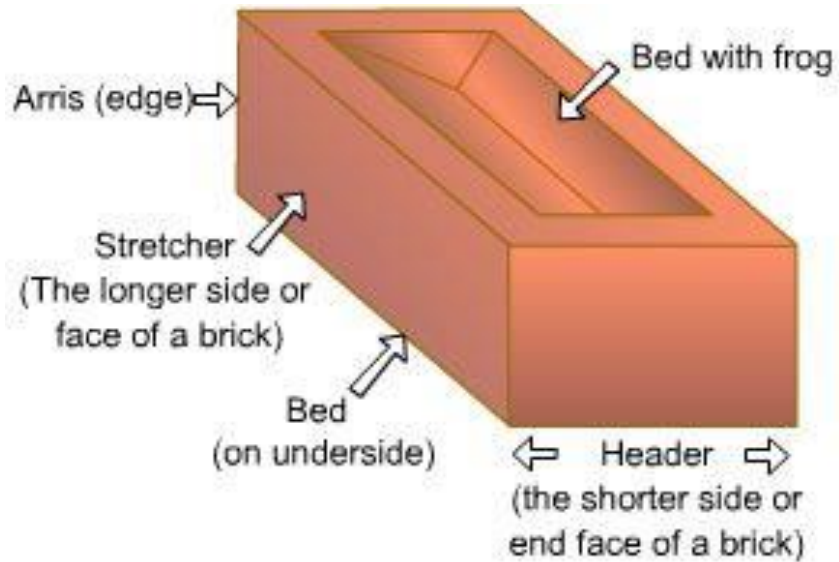
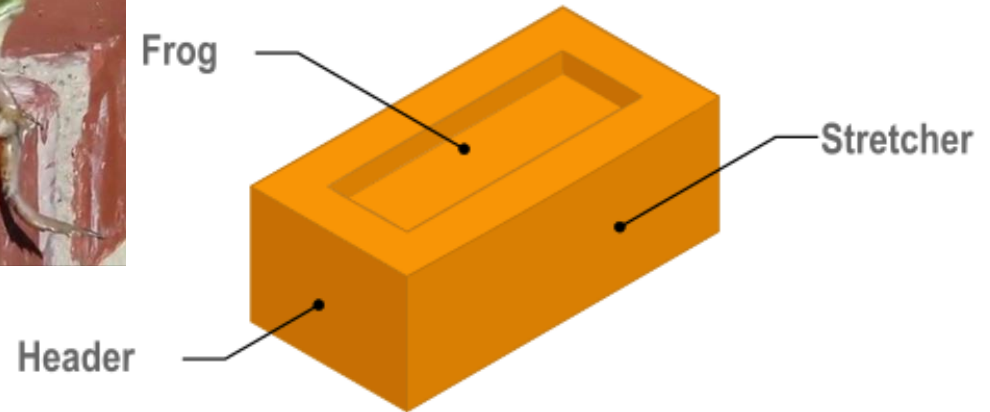
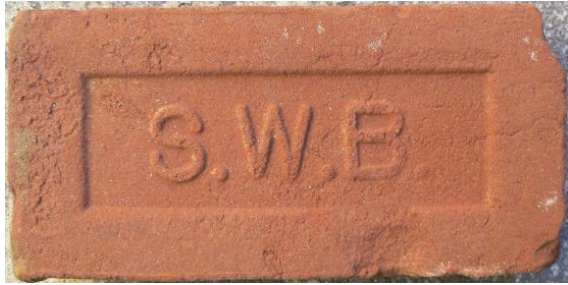
Characteristics of good bricks

- The bricks should have even surfaces, free from flaw or cracks and should have sharp well defined edges.
- The colour of the brick should be red or copper and uniform.
- On breaking the surface, it should show a bright homogeneous and compact surface free from voids.
- Should be so hard that no impression left when scratched with finger nails.
- Should produce clear ringing sound when struck against each other.

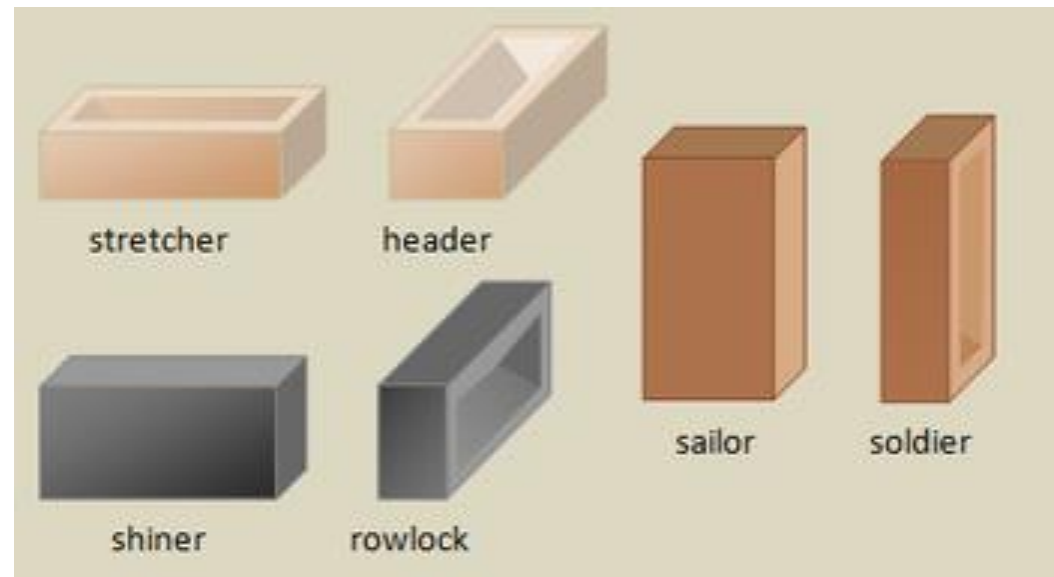
Characteristics of good bricks (contd...)

- No brick should have a crushing strength less than 3.5 N/mm^2 .
- Weight of the conventional brick $\approx 3 \text{ kgs}$
- Should not absorb water more than 15% of its weight.
- A brick soaked in water should not show any deposit of white salts after drying in shade.
- When dropped flat from a height of 1 m the brick should not break.
- Bricks should not conduct much heat and they should be sound proof.

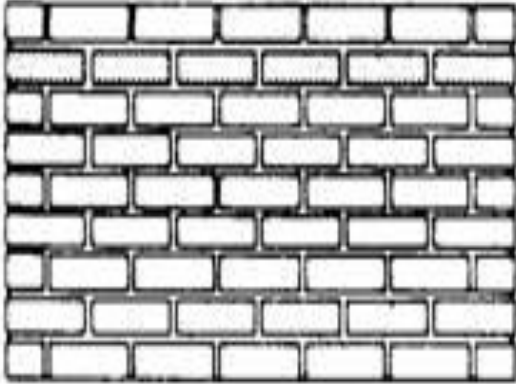
Parts of a brick



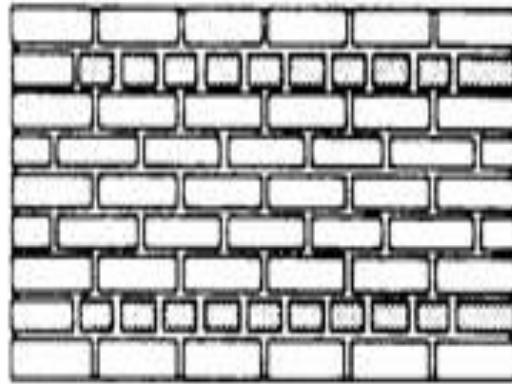
Faces of brick



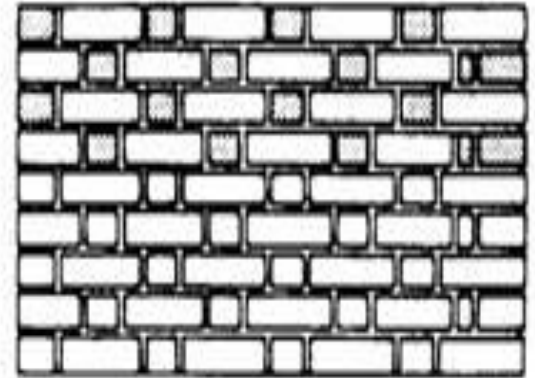
Different bonds in brick work



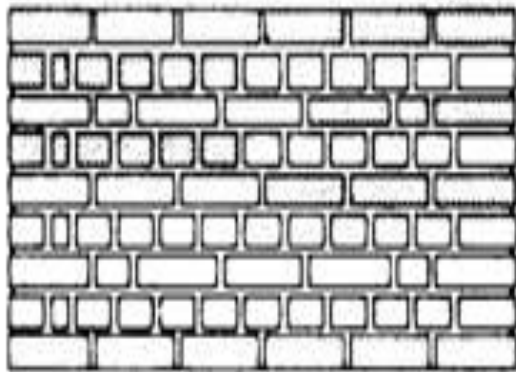
RUNNING



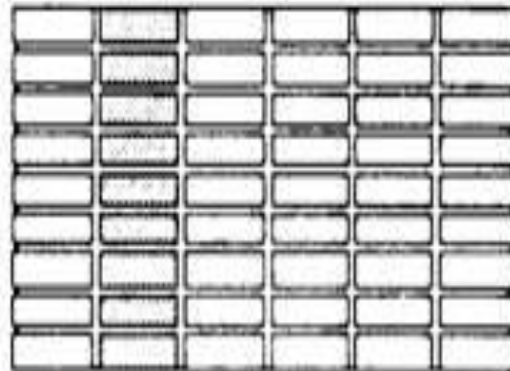
COMMON OR AMERICAN



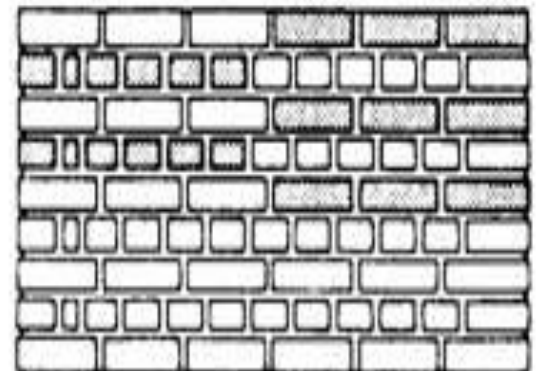
FLEMISH



ENGLISH



STACK



ENGLISH CROSS OR DUTCH

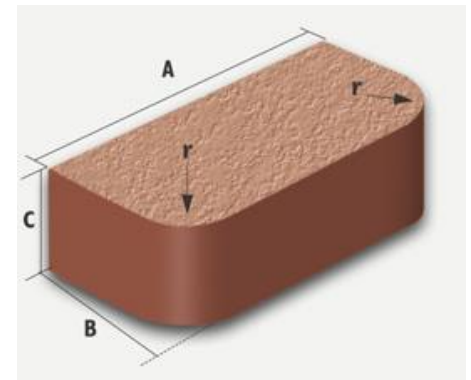
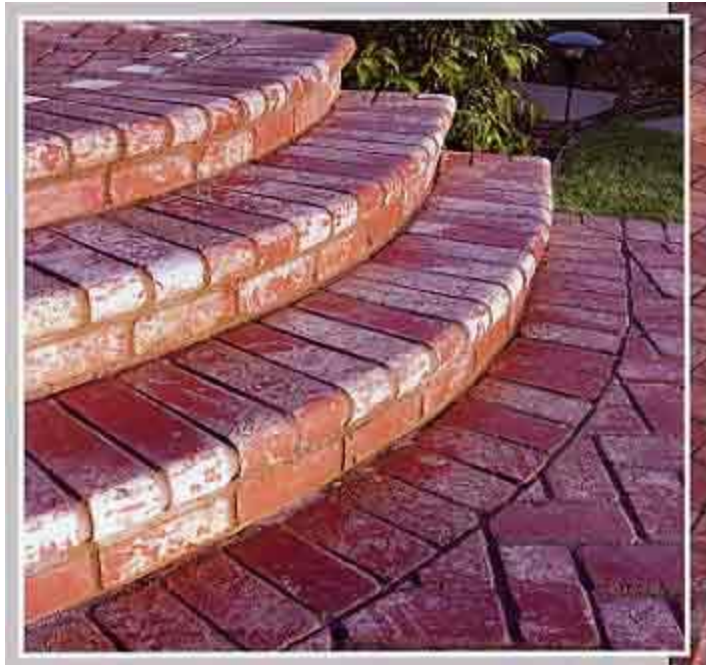
Brick types for special use

- **Acid brick or acid resistant brick** is specially made for masonry brick that is chemically resistant and thermally durable.
- Acid brick is created from high silica shale and fired at higher temperatures than those used for conventional brick.



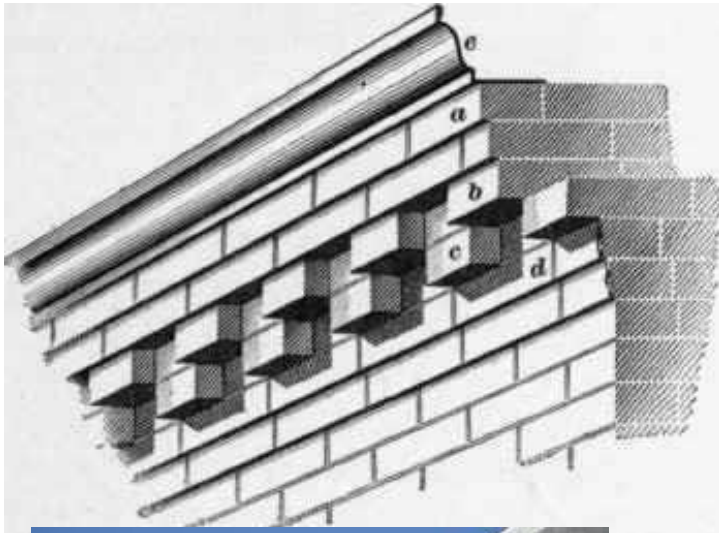
Bull-nosed brick

A bull-nose is a kind of brick that has one, some or all of its corners rounded off. Bull-nose bricks have been used to create a soft and attractive curve to the edge of the steps.



Cornice bricks

Several shapes and sizes of bricks when combined to form cornice are known as cornice bricks. These bricks are used to give better appearance.



Fire bricks or refractory brick

- A refractory brick is built primarily to withstand high temperature, but will also usually have a low thermal conductivity for greater energy efficiency. Usually dense firebricks are used in applications with extreme mechanical, chemical, or thermal stresses, such as the inside of a wood-fired kiln or a furnace, which is subject to abrasion from wood, fluxing from ash or slag, and high temperatures.
- Block of refractory ceramic material used in lining furnaces, kilns, fireboxes, and fireplaces.



Hollow Bricks

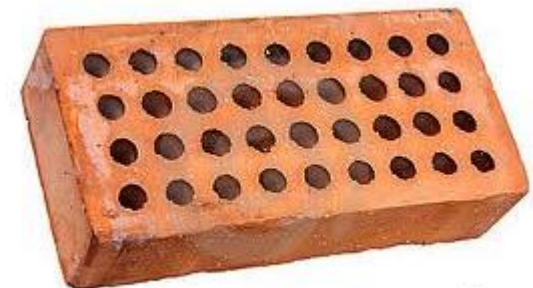
- These bricks are also known as cavity or cellular bricks.
- These bricks are not solid, but hollow.
- Their wall thickness may be 20 to 25 mm.
- They are light in weight and reduce transmission of heat, sound and dampness.
- They may be used for construction brick partitions.



Perforated bricks



- These bricks contain cylindrical holes throughout the thickness.
- These bricks are light in weight and they require less quantity of clay.
- These bricks are used in light weight construction like panel walls, partitions etc.



Plinth brick

- These bricks are used in plinth level and in sill level to create good appearance.



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